

7 (a) $\sin \theta = \lambda / b = (633 \times 10^{-9}) / (8.0 \times 10^{-5})$ [1]
 $\theta = 0.453^\circ$

$w = 2 \times 5.12 \times \tan (0.453^\circ)$ [1]
 $= 0.0810 \text{ m}$ (shown)

(if student use $\lambda D / a$, 0 immediately. Be aware....)

(b) (i) crests / light from second slit cancel troughs / light from first slit or destructive interference and reference to second slit or *owtte* [1]

(ii) $x = \lambda D / a = (633 \times 10^{-9})(5.12) / (2.40 \times 10^{-4}) = 0.0135 \text{ m}$ [1]

(iii) At these places are single slit diffraction minimums [1]
 No light to interfere [1]

